ment. Usually, an EMC/EMI filter does not have any capacitor to ground. This reduces the performance of the filter. As a result, a high-performance medical filter requires additional inductance to cancel out common mode noise.

Design of a custom filter solution depends on the level of insulation needed. The filter should not introduce any additional leakage current due to the parasitic effect in the system. However, an EMC filter with very low leakage current can be considered in less critical applications. Even with a filter that does not introduce any leakage current, overall leakage current for the whole system needs to be determined.

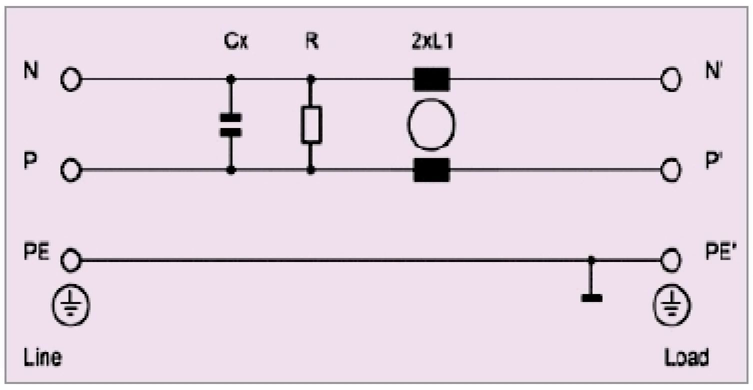

Fig. 2: B-type filter for no added leakage current (Credit: Schaffner Impulse)

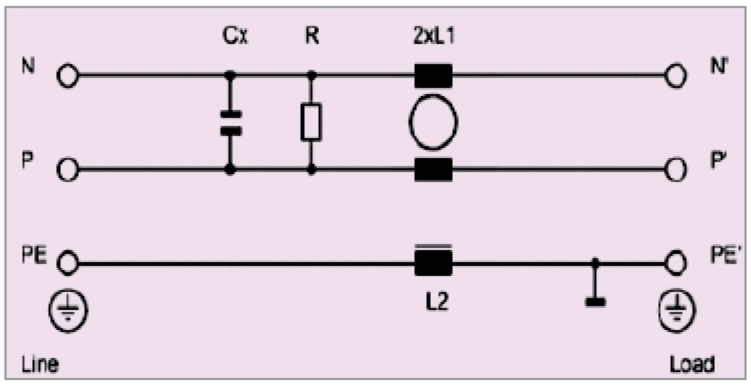

Fig. 3: Schematic of EB-type filter (Credit: Schaffner Impulse)

Schaffner Impulse provides FN92XX filters (B-type in particular) compliant with the requirements of EN/IEC 60601-1 standard regarding leakage current and voltage withstand. It offers standard solutions for different levels of leakage current requirements. Its B-type filters do not have any capacitance to PE and therefore do not add any leakage current to the system.

It also offers EB-type filter to compensate some of the lost performance as a result of no capacitance to PE. This type also increases immunity to electrical fast transients compared to B-type.

Electroplating to improve reliability and functionality of medical electronic devices

A metal coating protects against EMI without adding too much weight to medical equipment. It also strengthens digital equipment, extending its lifespan. Conductive materials, like copper, accomplish EMI protection by blocking electromagnetic emissions and then reflecting and absorbing them. An EMI shield can be made from a metal enclosure or metal coating. Using a conductive metal as a shield does not affect speed—it only blocks radio waves and other forms of EMI.

Electroplating can be used to coat parts with copper, nickel, gold, silver and other metals needed for EMI shielding. Sharretts Plating adheres to quality certifications such as ISO 13485 and ISO:9001 by providing required electroplating facilities. It makes sure that products are biocompatible by decreasing the risk of infection or rejection of implantable devices.

Conclusion

EMC filters for medical applications should fulfill the most stringent leakage current regulations, offer the best size versus performance and should be designed for typical usage in the medical environment.

EMI shielding is becoming more challenging because people are using medical devices also at homes, on the go and anywhere they need. According to Institute of Electrical and Electronics Engineers (IEEE), portable health devices tend to be more susceptible to EMI because these use low-power circuitry that can be more sensitive to electromagnetic fields. Medical devices are also becoming smaller and faster, and are being constructed with digital circuits. This poses as a challenge for device engineers because of faster digital speed and greater bandwidth requirement, making it more difficult to control electromagnetic emissions and their vulnerability. EFY